DEFENSIE
LA DÉFENSE

RMA
Royal Military Academy

DFR@RMA Call 2026

Final consensus report of research proposal SIGMA

1. Effectiveness

1.1. Project objectives

Rating: Excellent
Average score: 9,5/10
Explanatory comments for the evaluation: **Reviewer 1**: The project's objectives are perfectly described and strongly aligned. In addition, the project's objectives are clearly defined according to SMART criteria. **Reviewer 2**: The research objectives are well described; they are consistent with one another, define a consistent overall goal for the whole proposal and are perfectly SMART defined. The objectives are timely and very relevant given the serious threats that GNSS are currently experiencing. The proposal covers all domains for interference detection: signals, positions and networks, multi-constellation multi-frequency analysis. Moreover, it offers an end-to-end product, addressing from the data acquisition to the visualization.

2. Coherence between research objectives and methodology

2.1. Planned activities

Rating: Excellent
Average score: 10/10
Explanatory comments for the evaluation: **Reviewer 1**: It is convincingly presented in the proposal that the planned activities, as derived from the project objectives, can effectively achieve the project's objectives and result in the expected outputs/outcomes within the proposed time. **Reviewer 2**: The results framework and performance indicators are clearly defined. The tasks are described inside each workpackage in a consice but very clear and structured way. The planned activities are fully coherent with the project objectives. The tasks articulate the project in a very credible way, and they provide a very high degree of confidence that the objectives will be achieved. Moreover, the proposal puts forwards some solid ideas to maximise the impact of the project.

2.2. Methodology

Rating: Good
Average score:7/10
Explanatory comments for the evaluation: **Reviewer 1**: Overall, the proposed methodology and the use of data (i.e. RINEX data from existing GNSS reference stations) is well explained and appropriate to achieve the objectives and the expected outcomes. Regarding the implementation of coordinate-based anomaly detection algorithms, it is not mentioned in the proposal that the station coordinates would need to be calculated by the SIGMA platform itself, and how. This is also an essential element for comparisons at coordinate level after reprocessing with different combinations of constellations and/or frequencies.

Reviewer 2: The methodology lacks details. It is possible to understand how the project will be implemented because there are relevant discussions scattered along the proposal, but there is no specific section where the methodology is explained. It would have been appropriate to explain how many tasks will be done. For instance, there are many tasks involving algorithm development (e.g. ARIMA modelling, wavelet analysis, spatial correlation algorithms, etc.) where mentioning the source of the algorithms (i.e., whether they are already available in the group, or they are taken from a particular reference, or they are going to be developed in the project) and how the algorithms will be verified would have been appropriate.

3. Efficiency and adequacy of the work plan

3.1. Relation of the work packages to the proposal objectives

Rating: Excellent
Average score:10/10
Explanatory comments for the evaluation: **Reviewer 1**: - The work packages capture all aspects of the project's proposed objectives. - All work packages mention collaboration with the National Geographic Institute (NGI), which is important to the project. It is assumed that the NGI is committed to support the project, even though no NGI funding is requested through the project. **Reviewer 2**: The work plan outstandingly enables to apprehend all the objectives of the project with neither redundancies nor shortcomings. It is very clear that the workplan develops the objectives of the project in a systematic way.

3.2. Workplan

Rating: Good
Average score:7/10
Explanatory comments for the evaluation: **Reviewer 1**: - Due to the complexity of the project, it could be quite challenging for all tasks to be performed by one researcher. Involving a second researcher would also have fostered creative constructive collaboration between both researchers by idea-sharing and fresh perspectives during the project. That said, it is understood that the researcher can rely on the support of the project's promotor, among others to discuss any challenges arising during the course of the project. - In the Gannt chart, WPX is assigned zero hours. >> This is probably because WPX is led by the project's promotor, and no funding is required for WPX tasks. **Reviewer 2:** The work planning is elaborated in a well-thought manner (with the problems mentioned below), considering the efficiency, integration and synergy within the tasks. It leverages Belgium's dense GNSS network. The workplan presents a coherent and realistic roadmap. Deliverables are defined. The main problem is that the distribution of activities between the coordinator (RMA) and the other participants (NG!, BIPT, CPA, ACOS Transformation) is not defined. Several workpackages mention several participants, but there is no specific allocation of tasks or obligation to the different members.

3.3. Workload

Rating: Good
Average score: 7,5/10
Explanatory comments for the evaluation: **Reviewer 1**: The Gannt chart shows that, during some periods, there is a significant parallelisation of different tasks. This could be hard to manage by the researcher assigned to the project and could lead to difficulties in a fully successful completion of the project. **Reviewer 2**: According to the Gantt chart, it seems that the whole project will be done by a single person, working full-time for the project for 48 months. Although this is possibly feasible, it will be a very demanding assignment for that person. Moreover, it is highly surprising that neither the project leader nor other persons in the other participant institutions appear as allocation any effort to this project. This is probably an error in the proposal, and it cannot correspond to the reality, otherwise the project would not be feasible.

3.4. Risk assessment

Rating: Excellent
Average score: 9,5/10
Explanatory comments for the evaluation: **Reviewer 1:** The risks of each work package are properly assessed, and the proposed mitigation actions (e.g. solutions, alternatives) are appropriate to keep the risks under control. **Reviewer 2 :** Technical risks are adequately analysed in each workpackage. There is an additional risk related to the commitment of the other participant institutions (NGP, BPIT, etc.), which has not been mention. Given that the proposal does not give details about the involvement that these other institutions will have, there is the risk that they are not responsive and not contributing in the way that it is expected.

3.5. Budget assessment

Rating: Excellent
Average score: 10/10
Explanatory comments for the evaluation: **Reviewer 1** The budget addresses the needs of the project and aligns with the objectives and expected outcomes of the project. **Reviewer 2** The budget is modest, but commensurate with the goals of the project. It simply consists in the salary of one researcher during 48 months, funding to attend conferences, a monitor and a yearly subscription to software tool (with a very cheap cost). As a very minor comment, the subscription to an AI tool is mentioned, but the proposal hardly mentions AI and some more details of which AI/ML algorithms are going to be implemented would have been appreciated.

4. State of the Art and Innovative character

4.1. Knowledge of the state of the art

Rating: Excellent

Average score: 9,5 /10
Explanatory comments for the evaluation: **Reviewer 1**: The proposal demonstrates a deep overall understanding of the state of the art and includes references to key publications in the field of jamming and spoofing detection. **Reviewer 2**: The proposal contains a clear and to the point description of the state of the art. It focuses on the methods for interference monitoring that are very similar to the topic of the proposal. Given that interference monitoring is a subject that has attracted enormous interest among the GNSS community and numerous techniques have been proposed, a broader view on the state of the art would have been appreciated.

4.2. Position of the project with respect to the state of the art (innovativeness)

Rating: Good
Average score: 8 /10
Explanatory comments for the evaluation: **Reviewer 1**: The proposed concept is similar to projects that have already been undertaken or are under development in other countries, e.g. the interference monitoring and detection system implemented in the SWEPOS network in Sweden (as mentioned in the proposal). However, there is still great potential for innovation in the field of the relevant algorithms for GNSS data analysis at measurement level and at positioning level. This will be addressed by the proposed project which intends to introduce different technical innovations which will add value relative to the current state-of-the-art. This is nicely summarised in section 3.1 of the proposal. **Reviewer 2**: The technical innovation is good but not exceptional because the proposed techniques are described at a very high-level and they seem to be relatively standard and known in the community. For instance, at signal level only the signal strength and CN0 will be used, when many other techniques of signal quality monitoring exist and could have been appropriate for the project. In the domain of the position, integrity techniques developed to detect faults could have been used; on the contrary, it seems that the proposal relies on less sophisticated approaches based on characterizing the position errors. In any case, the proposal contains a section describing the ”Innovative character”, where some interesting contributions are mentioned.

5. Miscellaneous

Reviewer 1

Some technical aspects which could make the implementation of the proposed approach more challenging than anticipated are the following:

- GNSS receivers now implement jamming and spoofing countermeasures, meaning that indications of jamming/spoofing may not be present or may be less prominent in the RINEX log files.
- GNSS antennas used at CORS sites (e.g. choke ring antennas) are designed to reject signals from low elevation angles, meaning that they are less suitable for detecting interference coming from ground level. This will result in blind spots, even when using the dense CORS networks in Belgium.
- In case of spoofing, GNSS receivers may react in unpredictable ways, up to the point that logged data for all constellations and frequencies may become unusable, even if only one

frequency/constellation is spoofed. In general, it is considered particularly challenging to correctly identify spoofing based solely on RINEX log files.

- Regarding C/N0 monitoring, GPS Flex Power capability can introduce unexpected jumps, leading to false alarms if not accounted for.

Reviewer 2

The SIGMA proposal presents a very practical project with clearly defined objectives and a strong orientation toward operational outcomes. The expected results—particularly the GNSS interference detection platform—will be of clear utility to national stakeholders and are likely to have a significant impact on the broader PNT community.

The proposer demonstrates deep expertise in the topic and is exceptionally well positioned to lead the project successfully, based on a solid track record and strong institutional backing. While the degree of scientific innovation is moderate, the project represents a major advance in engineering and applied research, especially considering its emphasis on real-world implementation and national-scale deployment.

The requested resources are reasonable and proportionate to the goals, which reinforces the proposal's feasibility and cost-effectiveness. However, the roles and activities of other participants (e.g. NGI, BIPT, CPA) should be more clearly described, and the Gantt chart should explicitly reflect the involvement of personnel beyond the single hired researcher.